

Aditya Sugandhi

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SUMMARY

Senior Systems Engineer with 5+ years building production-grade distributed systems and cloud-native platforms using Python, Java, and Kubernetes. Deep expertise in reliability engineering, infrastructure-as-code (CloudFormation, Terraform), and AWS/cloud platforms (EKS, Lambda, MSK, DynamoDB). Proven track record leading technical projects end-to-end—from scoping and architecture to delivery—including monolith-to-microservices migrations, event-driven platforms processing 500k+ events/day, and observability infrastructure reducing MTTR by 60%. Strong technical mentor who drives team growth through knowledge sharing, design documentation, and cross-functional collaboration.

TECHNICAL SKILLS

- Systems Languages & Backend:** Python, Java (Spring Boot), Go (exposure), FastAPI, Microservices, Event-Driven Architecture, gRPC, REST/GraphQL APIs
- Distributed Systems & Reliability:** Kubernetes (EKS), Docker, Service Mesh, Circuit Breakers, Rate Limiting, Auto-Scaling, High Availability (99.5%+)
- Cloud Platforms & IaC:** AWS (EKS, Lambda, EC2, MSK, SQS/SNS, DynamoDB, S3, API Gateway, CloudWatch), CloudFormation, Terraform, Infrastructure-as-Code
- Data & Streaming:** PostgreSQL (Horizontal Sharding), DynamoDB, Redis, Elasticsearch, Kafka (MSK), Event Sourcing, Time-Series Databases, PySpark
- Observability & SRE:** Prometheus, Grafana, Kibana, CloudWatch, Distributed Tracing, SLA Monitoring, Alerting, 24/7 On-Call, MTTR Reduction
- DevOps & CI/CD:** Jenkins, GitHub Actions, Docker, Kubernetes, Argo, Blue-Green Deployments, Automated Testing (TDD/BDD, 85%+ coverage)
- Leadership & Collaboration:** Technical Mentorship (5+ engineers), Design Documentation, Cross-Functional Collaboration, Agile/Scrum, Project Scoping & Delivery

WORK EXPERIENCE

Software Engineer II
Florida State University

Nov 2024 – Present
Tallahassee, FL

- Led end-to-end design and delivery of distributed utility digital twin platform processing telemetry from 35k+ IoT devices across 172+ networks; owned scoping, architecture, work breakdown, and 6-month delivery timeline; built 37 REST/gRPC API endpoints with Python/FastAPI, achieving 99.2% accuracy and sub-500ms P99 latency.
- Architected fault-tolerant microservices on AWS using Lambda, SQS, SNS, and API Gateway with rate limiting, circuit breakers, and auto-scaling handling 10x traffic spikes; implemented PostgreSQL horizontal sharding and Redis distributed caching for high-throughput workloads.
- Built production observability stack with Prometheus and Grafana for 15+ microservices; designed operational dashboards with automated alerting on SLA violations, reducing MTTR by 60% through systematic troubleshooting and real-time performance metrics.
- Drove infrastructure-as-code adoption with CloudFormation and Docker containerization; established CI/CD pipelines (GitHub Actions, Jenkins) with 360+ unit/integration tests achieving 85%+ coverage and 99.5% deployment success rate.
- Mentored 3 junior engineers on distributed systems design, code quality standards, and TDD/BDD practices; collaborated with cross-functional teams (Product, Operations, QE) to identify high-impact problems and deliver solutions iteratively.

Software Engineer
Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024
Tallahassee, FL

- Independently architected and delivered distributed data pipeline platform on AWS EMR/EC2 processing 1TB+ datasets using Python (PySpark); owned technical scoping, implementation, and 4-month delivery; achieved +40% throughput and 99% job success rate through shuffle optimization and partitioning strategies.
- Built full-stack analytics platform with React.js frontend and Python/FastAPI backend; integrated Elasticsearch for real-time search across 50M+ records with sub-second query response serving 10K+ concurrent users.
- Deployed end-to-end monitoring infrastructure with Prometheus/Grafana and CloudWatch; automated drift detection, performance profiling, and SLA alerting reducing pipeline failures by 50% and enabling proactive capacity planning.
- Collaborated with stakeholders (facility managers, engineers, operations teams) to identify highest-impact problems; designed data integration workflows connecting enterprise systems and third-party APIs, enabling automated decision-making.

Software Engineer
Aspire Systems

Oct 2020 – Jul 2022
Chennai, India

- Led multi-month monolith-to-microservices migration on AWS EKS; owned architecture, technical scoping, work breakdown, and delivery across 3 teams; designed event-driven services in Java (Spring Boot) and Python; improved availability to 99.5%, reduced service coupling by 60%, and cut infrastructure costs by 20%.
- Architected real-time event streaming platform using Kafka (MSK), AWS SQS/SNS, and DynamoDB Streams with consumer groups, dead-letter queues, and automatic retry logic; processed 500k+ events/day with 10x spike handling during peak loads.
- Built distributed observability infrastructure with Kibana, Grafana, and CloudWatch for real-time system health monitoring; established 24/7 on-call rotation with PagerDuty, reducing MTTR by 60% through automated runbooks and systematic incident response.

- Deployed **Kubernetes auto-scaling infrastructure** with **Docker** and **CloudFormation IaC**; implemented **CodeBuild CI/CD pipelines** with blue-green deployments for rapid, safe releases across dev/staging/prod environments.
- **Mentored 5 engineers** on **Agile/Scrum** practices, microservices design patterns, and distributed systems architecture; established code review standards improving team velocity by **25%**; achieved **90% test coverage** with **TDD/BDD**.

PROJECTS

- **AI-Powered Resume Platform** — Built **full-stack SaaS** with **Next.js 14**, **Python/FastAPI**, and **AWS Lambda**. Integrated **LLM APIs (Groq)** for intelligent job matching and resume optimization. Deployed on **serverless architecture** with **DynamoDB** and **S3**. Implemented **observability** with CloudWatch metrics and distributed tracing. Tech: **React.js**, **Python**, **AWS Lambda**, **DynamoDB**, **AI/LLM integration**.
- **Worker-AI (Multi-Agent Platform)** — Architected **distributed task orchestration system** with **FastAPI** backend, **PostgreSQL**, **Redis** caching, and **React.js** frontend. Implemented **async event-driven** processing with **SQS**-style message queues and worker pools. Built **monitoring dashboards** for task metrics and system health. Containerized with **Docker Compose**. Tech: **Python**, **React.js**, **PostgreSQL**, **Redis**, **Docker**, **Event-Driven Architecture**.
- **Self-Hosted Cloud Infrastructure** — Provisioned **Linux server** with **Docker**-containerized services, **CI/CD pipelines**, and **automated deployments**. Hosts production apps including **Go microservices**, **Next.js frontends**, and **Python backends**. Implemented **monitoring** with Prometheus/Grafana and log aggregation. Tech: **Linux**, **Docker**, **Git**, **CI/CD**, **Observability**, **Security hardening**.

EDUCATION

Florida State University M.S. in Computer Science	Aug 2022 – May 2024 Tallahassee, FL
• Relevant Coursework: Distributed Systems, Advanced Database Systems, Cloud Computing, Data Mining	
SRM Institute of Science & Technology B.Tech. in Computer Science	Aug 2016 – May 2020 Chennai, India

CERTIFICATIONS & AWARDS

- AWS Solutions Architect Course (Udemy, 2024) — pursuing AWS Developer Associate certification
- Review Classification & False Feedback Detection (IJAST Publication)
- 2nd Place, SRM University Hackathon (2019)